

CS6187: VISION AND LANGUAGE

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Vision and Language

Subject Code

CS - Computer Science

Course Number

6187

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

CS5187 Vision and Image or
CS5286 Algorithms and Techniques for Web Searching or
CS5487 Machine Learning: Principles and Practice

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course introduces algorithms and techniques for integration of computer vision and natural language processing for innovative applications, such as robot dialog system, image/video captioning, cross-media multimedia search and question-answering. The course will discuss the latest technologies in bridging the gap between vision and language, with topics ranging from machine translation, feature extraction and learning, to design of deep neural network architecture.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if DEC-A1 app.)		DEC-A2	DEC-A3
1	Describe techniques in visual feature extraction, learning and representation.			x	
2	Explain the process of natural language processing, ontology construction and knowledge inferencing.			x	
3	Perform critical assessment on the effectiveness of rule-based, statistical-based and machine learning based translation models.		x		
4	Apply computer vision and natural language processing for innovative applications.		x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Students will engage on topics about the introduction of computer vision natural language processing machine translation, and related real-world applications such as image/video captioning and robot dialog system.	1, 2, 3	2 hours/ week

2	Tutorial	Students will work on a different problem set each week during the tutorial sessions, through which they can discover the differences and commonalities in vision, speech and text, and the solutions to bridge the gap between different domains. They will also be invited to present their solutions, and the class will be encouraged to provide comments.	1, 2, 3	1 hour/ week
3	Assignment	The students will implement basic vision and language algorithms for processing different modalities, and translation models in modelling or learning the embedded relationship between different modalities. In this way, students can observe the characteristics and perform critical assessment of different machine translation models.	3	
4	Project	The students will create a new system design or solution to solve a real-world problem. The students will apply the principles they have learnt from the course for innovative application.	4	

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- for nil entry)	Allow Use of GenAI?
1	Assignments	1, 2, 3	30	Expect to have at least two assignments.	Yes
2	Project	4	40	-	Yes

Continuous Assessment (%)

70

Examination (%)

30

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

Assessment Rubrics (AR)

Assessment Task

Assignment (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability to implement and assess the effectiveness of different algorithms and techniques.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability and creativity in applying appropriate algorithms and techniques for real-world applications that require integration of vision and language.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The extent to which the students can understand the differences and commonalities between different modalities and ability to perform appropriate algorithms and techniques for integration of them, and evaluate their performances.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Assignment (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability to implement and assess the effectiveness of different algorithms and techniques.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability and creativity in applying appropriate algorithms and techniques for real-world applications that require integration of vision and language.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The extent to which the students can understand the differences and commonalities between different modalities and ability to perform appropriate algorithms and techniques for integration of them, and evaluate their performances.

Excellent

(A+, A, A-) High

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(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal levels

Part III Other Information

Keyword Syllabus

Visual understanding, natural language processing, speech transcription, knowledge graph representation and reasoning, canonical correlation analysis, feature extraction and learning, deep neural network, image / video tagging and captioning, zero-shot learning, question-answering.

Reading List**Compulsory Readings**

Title	
1	Nil

Additional Readings

Title	
1	Yoshua Bengio, Ian Goodfellow, Aaron Courville, Deep Learning MIT Press, 2014.
2	Richard Szeliski, Computer Vision: Algorithms and Applications Springer; 2011.
3	Daniel Jurafsky, James H. Martin, Speech and Language Processing Prentice Hall, 2009.
4	Philipp Koehn, Statistical Machine Translation Cambridge University Press, 2010.